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Inhibition of CXCL10 and IFN- γ ameliorates myocarditis in preclinical models of SARS-CoV-2 mRNA vaccination

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Messenger RNA (mRNA) vaccines against SARS-CoV-2 are highly effective and were instrumental in curbing the COVID-19 pandemic. However, rare cases of noninfective myocarditis, particularly in young males and typically after the second dose, have been observed. Here, we explore the mediators of this myocarditis to better understand and to enhance the safety of future mRNA vaccines. Through analysis of human plasma data and in vitro experiments with human macrophages and T cells, we identified increased C-X-C motif chemokine ligand 10 (CXCL10) and interferon- γ (IFN- γ) after exposure to BNT162b2 (Pfizer) or mRNA-1273 (Moderna). Neutralization of CXCL10 and IFN- γ during the second dose (21 days after the first dose) reduced vaccine-induced cardiac injury in mice. Neutralization also reduced cardiac stress markers such as the release of N-terminal pro-B-type natriuretic peptide (NT-proBNP) and expression of inflammatory genes in human induced pluripotent stem cell (iPSC)-derived cardiac spheroids. When exposed to these cytokines in vitro, human iPSC-derived cardiomyocytes (iPSC-CMs) exhibited impaired contractility, arrhythmogenicity, and proinflammatory gene expression patterns. Genistein, a phytoestrogen implicated in reducing cardiovascular inflammation, mitigated these effects in iPSC-CMs. In mice exposed to these cytokines or receiving BNT162b2 vaccination, genistein treatment reduced cardiac injury markers and attenuated infiltration of neutrophils and macrophages into the heart. These findings implicate CXCL10–IFN- γ signaling as a contributor to myocardial injury in experimental models of mRNA vaccination and indicate that pharmacologic modulation, such as with genistein, may mitigate cytokine-driven injury.

INTRODUCTION

By leveraging nucleoside-modification technologies that increase stability and minimize immunogenicity (1, 2), mRNA vaccines have demonstrated exceptional efficacy in preventing symptomatic COVID-19 and altered the course of the pandemic (3–6). The mRNA vaccines for COVID-19 have also shown a robust safety profile (3–6). Given the relatively new nature of this technology, continuous safety monitoring has been essential. Among the observed adverse effects of mRNA vaccines, myocarditis has been reported as a rare but concerning cardiac complication (7–13). Vaccine-associated myocarditis has been reported primarily in young males (male, 79 to 94%; average age, 15.9 years old), particularly after the second dose (8, 11, 13). The incidence post first dose is ~7.2 cases per million vaccine doses, increasing to 31.3 cases per million doses post second dose, with higher incidence observed in males under 30 years of age [59.7 cases per million doses; 95% confidence interval (CI), 29.8 to 119.4] (12). Symptoms typically appear ~2.6 days postvaccination (95% CI, 1.9 to 3.3 days), presenting as chest pain, shortness of breath, fever, and palpitations (13). In clinical tests, the majority of cases (>84%) present with elevated cardiac

troponin I (cTnI), which indicates myocardial injury, and more than half of cases show ST-segment changes in electrocardiograms (EKGs) (13). Additionally, about 87% of cases show late gadolinium enhancement in cardiac magnetic resonance imaging (MRI), indicating fibrosis after myocardial inflammation (13). Although most maintain normal left ventricular function and recover with favorable outcomes (9–14), a few cases with more critical outcomes have also been reported (10, 13). The distinct characteristics of mRNA vaccine-associated myocarditis (such as the absence of pathogen infection, rapid onset postexposure, and prompt recovery timelines) suggest a unique underlying mechanism diverging from typical viral myocarditis.

The scope of mRNA technology is expected to expand beyond COVID-19 vaccines across various biomedical fields. Potential applications include vaccines for other infectious diseases, anticancer immunotherapies, gene editing, and regenerative medicine (1, 2, 15, 16). Therefore, understanding the mechanisms behind such rare adverse effects is crucial to improving the safety of future mRNA-based vaccines and therapies.

This study aimed to elucidate the mechanisms behind myocardial injury observed after COVID-19 mRNA vaccination by using multifaceted experimental models including human induced pluripotent stem cells (iPSCs) and mouse models, complemented by data analysis of human plasma samples. Additionally, we evaluate the effectiveness of genistein, an anti-inflammatory phytoestrogen (17), as a potential countermeasure against this adverse effect.

RESULTS

CXCL10 and IFN- γ are up-regulated after COVID-19 mRNA vaccination

Clinical studies showed that the highest incidence of vaccine-associated myocarditis was observed post second dose in young

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males (9, 12, 13). Correspondingly, in the Vaccine Adverse Event Reporting System (VAERS) data, myocarditis cases are mainly reported in young males, typically occurring 1 to 3 days post second dose (Fig. 1, A to C). Cases have also been reported after the first dose, albeit less frequently. This pattern of very early onset (with peaks as early as day 1 postvaccination) suggests a pathophysiological mechanism distinct from that of viral myocarditis (18). Extending these epidemiologic observations, studies analyzing blood samples from patients with vaccine-associated myocarditis reported elevations of circulating cytokines and activation of innate immunity pathways without marked changes in humoral responses (19, 20). Furthermore, the 1- to 3-day postvaccination window aligns with the reported activation period of the innate immune system in both humans and mice (21, 22). Together, these observations—the early onset of myocarditis within 1 to 3 days of vaccination, cytokine elevation in clinical samples, and the overlap with the innate immune activation period—prompted us to examine whether early cytokine induction contributes to myocardial injury after mRNA vaccination.

To identify cytokines potentially involved in the development of myocardial injury postvaccination, we first analyzed human data from two key studies. The first was a time-course analysis of plasma cytokine amounts in healthy individuals after vaccination with the BNT162b2 (Pfizer mRNA vaccine) (22). The second study was on vaccine-associated myocarditis versus young vaccinated controls (19). Among the 27 cytokines detected in both studies, we focused on C-X-C motif chemokine ligand 10 (CXCL10) and its upstream cytokine, interferon- γ (IFN- γ), for two main reasons. First, in the time-course study (22), these cytokines were the most elevated on days 1 to 2 postvaccination (after both the first and second doses), with a greater increase on day 1 after the second dose; this timing aligns with the early onset of vaccine-associated myocarditis (typically 1 to 3 days post second dose) (Fig. 1C and fig. S1, A to D). Second, in the comparative study (19), the average amounts of CXCL10 and IFN- γ were even more elevated in patients with vaccine-associated myocarditis than in vaccinated controls (fig. S1E).

Next, we assessed cytokine production by myeloid cells, the primary responders to mRNA vaccines in vitro (1). Human iPSC-derived macrophages (iPSC-MACs) (23, 24) were treated with phosphate-buffered saline (PBS; control), polyethylene glycol 2000 (PEG2000; vehicle), BNT162b2 (Pfizer), or mRNA-1273 (Moderna) (fig. S2A). Two concentrations were set for each vaccine: Pfizer low (0.6 μ g/ml), Moderna low (2 μ g/ml), Pfizer high (3 μ g/ml), and Moderna high (10 μ g/ml). These concentrations reflect the relative mRNA content of the clinical formulations (Pfizer, 30 μ g; and Moderna, 100 μ g) (1). A multiplexed cytokine array at 24 hours posttreatment showed a dose-dependent elevation of CXCL10 with both Pfizer and Moderna vaccines and an increase of IFN- γ with Pfizer high (Fig. 1D). Other IFN family members involved in antiviral responses, such as IFN- λ and IFN- α 2, were also induced by both vaccines (fig. S2B). Generic inflammatory cytokines such as tumor necrosis factor- α (TNF- α) and interleukin-10 (IL-10) remained minimally induced (Fig. 1D and fig. S2B), consistent with their plasma amounts detected in humans (fig. S1, A to E) (19, 22), suggesting a selective up-regulation of the CXCL10-IFN- γ pathway in response to mRNA vaccination.

RNA sequencing (RNA-seq) of iPSC-MACs 24 hours posttreatment revealed distinct transcriptomic profiles (Fig. 1E). The two vaccines shared 604 up-regulated genes, including IFN- γ -responsive chemokines *CXCL9*, *CXCL10*, and *CXCL11* (Fig. 1, F and G),

highlighting the crucial role of IFN- γ signaling. Gene ontology (GO) analysis revealed enriched pathways related to external nucleotide response, cytokine-mediated signaling, and T cell activation in response to both vaccines (Fig. 1, H and I; and fig. S3, A to C). Furthermore, comparison of our iPSC-MAC RNA-seq data with published monocyte data from vaccinated individuals (22) demonstrated strong transcriptomic concordance, indicating that our iPSC-MACs accurately mimic the vaccine responses of myeloid cells in humans (fig. S4, A and B).

We also investigated the cellular sources of CXCL10 and IFN- γ . iPSC-MACs treated with BNT162b2 exhibited increased *CXCL10* and *IFNG* mRNA expression at 24 hours (Fig. 1J), which was consistent with protein-level changes observed above (Fig. 1D). Coculture experiments demonstrated that, although vaccine-treated macrophages secreted low amounts of IFN- γ , coculturing these macrophages with T cells or culturing T cells with conditioned medium from these macrophages amplified IFN- γ production, indicating that T cells are the primary source of IFN- γ (Fig. 1, K and M). In contrast, direct treatment of T cells with BNT162b2 or with conditioned medium from vaccine-treated macrophages did not induce CXCL10 production, confirming that CXCL10 was produced by macrophages (Fig. 1, K and L). These findings are consistent with the single-cell RNA-seq (scRNA-seq) dataset of human peripheral blood mononuclear cells (PBMCs) (22), which shows that *CXCL10* mRNA was predominantly expressed in monocytes, whereas *IFNG* mRNA was mainly expressed in T cells and NK cells postvaccination (fig. S4, C and D).

Moreover, conditioned medium from vaccinated macrophages activated T cells, as demonstrated by increased extracellular signal-regulated kinase (ERK) phosphorylation (fig. S5, A to C), and promoted T cell migration in a transwell assay (fig. S5, D and E). These effects were blocked by AMG487 (an inhibitor of CXCL10's receptor, CXCR3) (fig. S5B, C, and E), indicating the interaction between macrophages and T cells through CXCL10. This interaction was further corroborated by reanalyses of the two scRNA-seq datasets of human PBMCs postvaccination (19, 22), which indicated that T cell activation is the top enriched GO in monocytes from patients with vaccine-associated myocarditis (fig. S4, E and F).

CXCL10 and IFN- γ contribute to cardiac injury in mouse and spheroid models of mRNA vaccination

To determine whether CXCL10 and IFN- γ contribute causally to postvaccination cardiac injury, we first established a mouse model and measured serum cTnI, a clinical marker of myocardial injury (Fig. 2A). Wild-type BALB/c mice received BNT162b2 intramuscularly on days 0 and 21 (mimicking the standard human vaccination schedule) at doses ranging from 0.2 μ g (body-surface-area-adjusted human-equivalent dose) to 5 μ g. The 0.2- μ g dose produced only a slight, clinically unremarkable increase in cTnI, whereas higher doses (2 and 5 μ g) caused a marked increase, peaking 6 to 10 hours after the second injection (Fig. 2B). Despite elevated cTnI, left-ventricular ejection fraction remained normal, consistent with clinical observations (fig. S6, A and B). Serum cTnI concentrations did not differ significantly between male and female mice at 7 or 24 hours postvaccination (fig. S6C). To confirm that the cardiac effects were not strain-specific, we repeated the protocol in C57BL/6 mice. The 2- μ g dose again triggered a robust cTnI release and up-regulated inflammatory gene expression in the heart (fig. S6, D to F). To rule out a contribution from the lipid carrier, mice injected with an equivalent

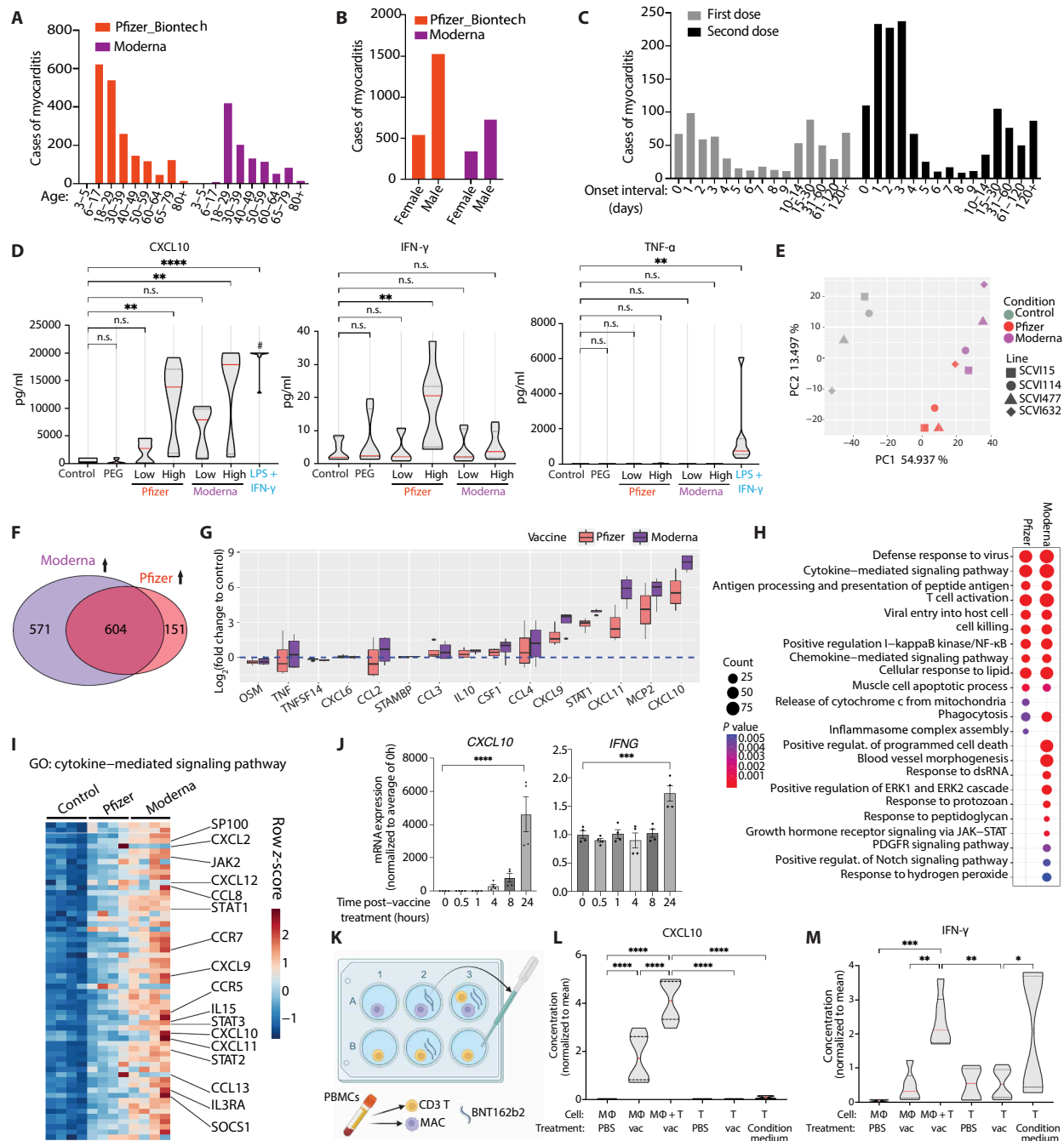


Fig. 1. COVID-19 mRNA vaccination induces CXCL10 and IFN-γ up-regulation in immune cells. (A to C) VAERS-reported myocarditis cases after BNT162b2 or mRNA-1273 vaccination, stratified by age (A), sex (B), and day of onset (C). (D) Cytokine concentrations in supernatants of iPSC-MACs 24 hours after treatment with PBS, PEG2000 (5 μg/ml), BNT162b2 [low (0.6 μg/ml) or high (3 μg/ml)], mRNA-1273 [low (2 μg/ml) or high (10 μg/ml)], or positive control [lipopolysaccharide (LPS; 10 ng/ml) and IFN-γ (20 ng/ml)]; data from seven iPSC lines. “#” indicates upper detection limit. (E) Principal component (PC) analysis of control versus vaccine-treated iPSC-MACs ($n = 4$ lines). (F) Differentially expressed genes (DEGs) up-regulated by vaccines (false discovery rate < 0.05 ; fold change > 2). Full DEG lists are in data file S1. (G) Fold change of cytokine genes in vaccine-treated iPSC-MACs. $n = 4$ iPSC lines. (H) GO enrichment of up-regulated DEGs ($q < 0.05$). Full GO lists are in data file S2. dsRNA, double-stranded RNA; ERK, extracellular signal-regulated kinase; JAK, Janus kinase; PDGFR, platelet-derived growth factor receptor. (I) Heatmap shows expression of genes within the selected GO across different groups of iPSC-MACs. Color represents row z-score. (J) CXCL10 and IFNG mRNA expression in BNT162b2-treated iPSC-MACs (3 μg/ml); four independent experiments. (K) Schematic of coculture experiment. CD3 T cells and macrophages were isolated from PBMCs from the same donor and cultured individually or together as shown. Supernatants were collected after 24 hours of treatment with BNT162b2 (3 μg/ml). For the B3 condition (T and condition medium), supernatants were collected after 24 hours of treatment with the condition medium from A2 (macrophage and BNT162b2) at a concentration of 50% (v/v). Artwork created using BioRender.com. (L) CXCL10 and (M) IFN-γ secretion 24 hours after BNT162b2 treatment (3 μg/ml); $n = 6$ (2 PBMC donors $\times$ 3 independent experiments). For multiple comparisons in (D), (J), (L), and (M), statistical significance was determined using ANOVA with Holm-Šidák's post hoc test. Error bars represent $\pm$ SEM. Statistical significance is defined as not significant (n.s.), $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$.

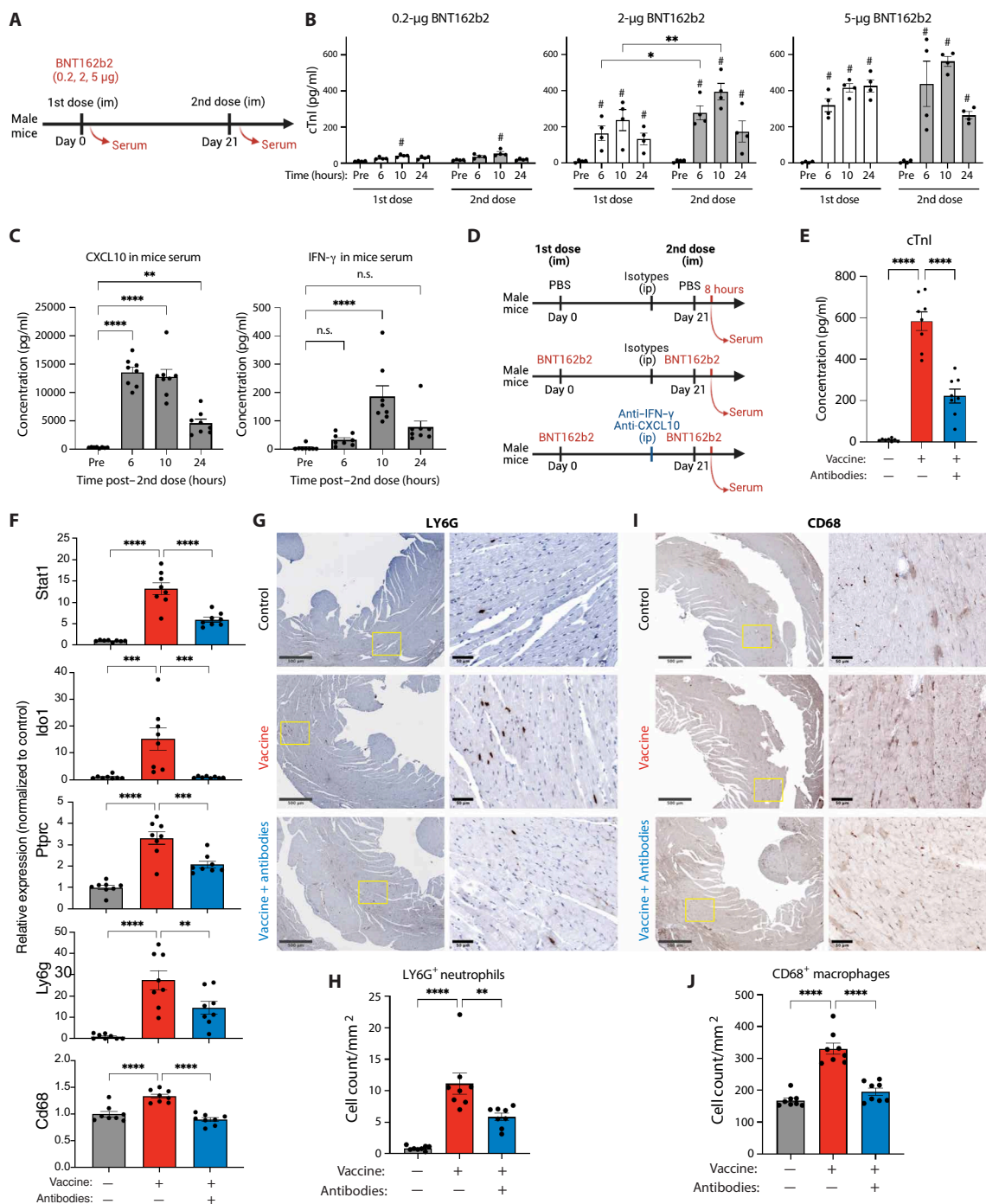


Fig. 2. Targeting CXCL10 and IFN- γ attenuates vaccine-induced myocardial injury in mice. (A) Optimization of the mouse model using two doses of BNT162b2. (B) Serum cTnI after BNT162b2 vaccination (0.2 to 5 μg) in BALB/c mice, measured at indicated time points (n = 4 per group). # represents $P < 0.05$ compared with the pretreatment group within the same dose. (C) Serum CXCL10 and IFN- γ concentrations after 2-μg BNT162b2 vaccination. (D) Schematic of neutralization strategy: Anti-CXCL10 and anti-IFN- γ antibodies were administered before the second vaccine dose on day 21 (n = 8 mice per group). ip, intraperitoneal. (E) Serum cTnI measured 8 hours after second vaccination. n = 8 mice per group. (F) Cardiac expression of IFN- γ -responsive genes (*Stat1* and *Ido1*) and leukocyte markers (*Ptprc*, *Ly6g*, and *Cd68*). Data were normalized to the mean of the control group. (G) Immunohistochemistry showing LY6G⁺ neutrophil infiltration in vaccinated hearts, attenuated by cytokine neutralization. Enlarged insets from representative areas (yellow boxes) are shown at right. Scale bars, 500 μm (left) and 50 μm (right). (H) Quantification of LY6G⁺ cells. n = 8 mice. Ten field of views were quantified for each section per mouse. (I) Immunohistochemistry showing CD68⁺ macrophage infiltration in vaccinated hearts, attenuated by cytokine neutralization. Enlarged insets from representative areas (yellow boxes) are shown at right. Scale bars, 500 μm (left) and 50 μm (right). (J) Quantification of CD68⁺ cells. n = 8 mice. Ten field of views were quantified for each section per mouse. For multiple comparisons in (B), (C), (E), (F), (H), and (J), statistical significance was determined using ANOVA with Holm-Šidák's post hoc test. Error bars represent $\pm$ SEM. Statistical significance is defined as not significant (n.s.), * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

amount of PEG2000 vehicle showed no increase in cTnI or inflammatory genes, aside from a modest rise in *Cd68* mRNA (fig. S6, E and F). Collectively, these data show that the 2- μ g BNT162b2 regimen reliably modeled vaccine-induced cardiac injury, preserved the second-dose predominance seen in patients, and produced concurrent rises in serum CXCL10 and IFN- γ (Fig. 2C). We therefore selected the 2- μ g dose for subsequent experiments.

To investigate whether CXCL10 and IFN- γ drive this vaccine-induced cardiac injury, we administered neutralizing antibodies 8 hours before the second dose. Mice were divided into three groups: unvaccinated controls, vaccine alone, and vaccine plus CXCL10/IFN- γ neutralization (Fig. 2D). Blocking both cytokines reduced the vaccine-induced increase in serum cTnI measured 8 hours after the second dose (Fig. 2E). Neutralization also lowered cardiac mRNA amounts of IFN- γ -responsive genes, such as signal transducer and activator of transcription 1 (*Stat1*) and indoleamine 2,3-dioxygenase 1 (*Ido1*) (Fig. 2F), and leukocyte markers, such as protein tyrosine phosphatase receptor type C (*Ptpcr*) (pan-leukocyte marker), lymphocyte antigen 6 complex, locus G (*Ly6g*) (neutrophil marker), and cluster of differentiation 68 (*Cd68*) (macrophage marker) (Fig. 2F). Immunohistochemistry confirmed that vaccination led to marked neutrophil and macrophage infiltration into cardiac tissue, which was mitigated by CXCL10/IFN- γ neutralization (Fig. 2, G to J). In contrast, CD3⁺ T cell infiltration also increased after vaccination but was not altered by cytokine neutralization (fig. S7A and B).

CXCL10/IFN- γ neutralization did not broadly suppress the normal vaccine response. Serum profiling showed no reduction in antiviral cytokines other than C-C motif chemokine ligand 2 (CCL2) (fig. S7C), and anti-SARS-CoV-2 Spike immunoglobulin G (IgG) titers remained ~80% of those in vaccine-only mice (fig. S7D). Together, these data indicate that CXCL10 and IFN- γ were mediators of cardiac injury after mRNA vaccination, and their neutralization largely preserved systemic immune activation and humoral immunity.

Consistent with the immune cell infiltration in vivo, in vitro experiments demonstrated that the conditioned medium from vaccinated immune cells increased cell-surface intercellular adhesion molecule 1 (ICAM-1) and vascular cell adhesion molecule 1 (VCAM-1) on iPSC-derived endothelial cells (iPSC-ECs), and this effect was blocked by CXCL10/IFN- γ inhibition (fig. S8, A to C). The same conditioned medium enhanced adhesion of THP-1 and Jurkat cells to endothelial cells, and this increase was mitigated by cytokine inhibition (fig. S8, D to G).

To further investigate the impact of CXCL10 and IFN- γ on cardiac injury, we performed in vitro assays using cardiac spheroids composed of iPSC-derived cardiomyocytes (iPSC-CMs) (25, 26), iPSC-MACs, and iPSC-ECs (Fig. 3, A to D). Because our data indicated that CXCL10 was primarily produced by macrophages and IFN- γ by T cells (Fig. 1, K to M), we treated cardiac spheroids with conditioned medium from vaccine-stimulated macrophages and T cells, with or without cytokine inhibitors (anti-IFN- γ neutralizing antibody and AMG487). The conditioned medium from vaccinated immune cells increased N-terminal pro-B-type natriuretic peptide (NT-proBNP) release into the supernatant, a marker of cardiac stress, which was rescued by cytokine inhibitors (Fig. 3E). Similarly, contractile functions of cardiac spheroids, including beating rate, contraction velocity, and relaxation velocity, were impaired by the conditioned medium but partially restored by cytokine inhibition, particularly for beating rate and relaxation velocity (Fig. 3F and movie S1). To assess cell-type-specific responses, we sorted cardiac

spheroids into cardiomyocytes, macrophages, and endothelial cells for gene expression analysis of IFN- γ -responsive genes. Conditioned medium up-regulated *STAT1*, *IDO1*, and *CXCL10* expression across all three cell types (Fig. 3G). The cytokine inhibitors effectively reversed *STAT1* up-regulation in endothelial cells and macrophages, *IDO1* in cardiomyocytes and macrophages, and *CXCL10* in all three cell types (Fig. 3G). Collectively, these in vitro findings further supported our in vivo results, demonstrating a contributory role of CXCL10 and IFN- γ in postvaccination cardiac injury models.

CXCL10 and IFN- γ impair cardiomyocyte function in vitro

To investigate the impact of CXCL10 and IFN- γ on cardiomyocytes, we next treated human iPSC-CMs with each cytokine alone or in combination. CXCL10 alone did not alter the expression of natriuretic peptide B (*NPPB*) (cardiac stress marker) or IFN- γ -responsive genes, whereas IFN- γ alone up-regulated these genes (fig. S9). The combined treatment led to a modestly greater response than IFN- γ alone in IFN- γ -responsive genes (fig. S9), suggesting an additive interaction with IFN- γ as the primary mediator. On the basis of these results, we used the cytokine cocktail in subsequent experiments.

iPSC-CMs were treated with the cytokine cocktail at two concentrations: a “low” concentration [CXCL10 (1 ng/ml) and IFN- γ (0.5 ng/ml)] and a “high” concentration [CXCL10 (10 ng/ml) and IFN- γ (5 ng/ml)]. These concentrations were estimated using published Olink proteomics data (22) and cytokine array data (19) to represent the peak cytokine amounts postvaccination in healthy individuals and in patients with myocarditis, respectively. The high concentration reduced contraction velocity, relaxation velocity, and contraction deformation distance in iPSC-CMs. The low concentration also adversely affected the contractility of iPSC-CMs, albeit to a lesser extent (Fig. 4, A to C). The release of NT-proBNP was elevated at 48 hours posttreatment in a dose-dependent manner (Fig. 4D).

Electrophysiological phenotypes were further analyzed using multielectrode arrays (MEAs). The high concentration led to an increase in arrhythmia-like events (Fig. 4E). Prolongation of corrected field potential duration (cFPD) and beating period (BP) was observed at 48 hours after the treatment with both low and high concentrations, indicating an increased susceptibility to arrhythmic events (Fig. 4, F to I).

RNA-seq showed that both low and high cytokine concentrations triggered substantial transcriptomic changes in iPSC-CMs (Fig. 4J and fig. S10A). The high concentration up-regulated 473 genes compared with control, mainly related to inflammatory response, cytotoxicity, muscle hypertrophy, and mitogen-activated protein kinase (MAPK) pathways (fig. S10, B to G), mimicking myocarditis-like gene expression patterns. Of note, we found that genes involved in the proteasomal pathway and immunoproteasome activation were up-regulated, highlighting a unique feature of cytokine-induced cardiac damage (Fig. 4J and figs. S10, H and I, and S11A) (27–29).

Genistein protects cardiomyocytes from cytokine-induced injury in vitro

Given the higher incidence of vaccine-associated myocarditis in males reported in clinical studies (9–13) and the well-documented cardiovascular benefits of female hormones (30, 31), we tested whether estrogens could reduce cytokine-induced myocardial injury. We therefore examined estradiol (E2) and genistein, a phytoestrogen with anti-inflammatory activity as demonstrated in our prior work (17). iPSC-CMs were pretreated with E2 or genistein for 48 hours

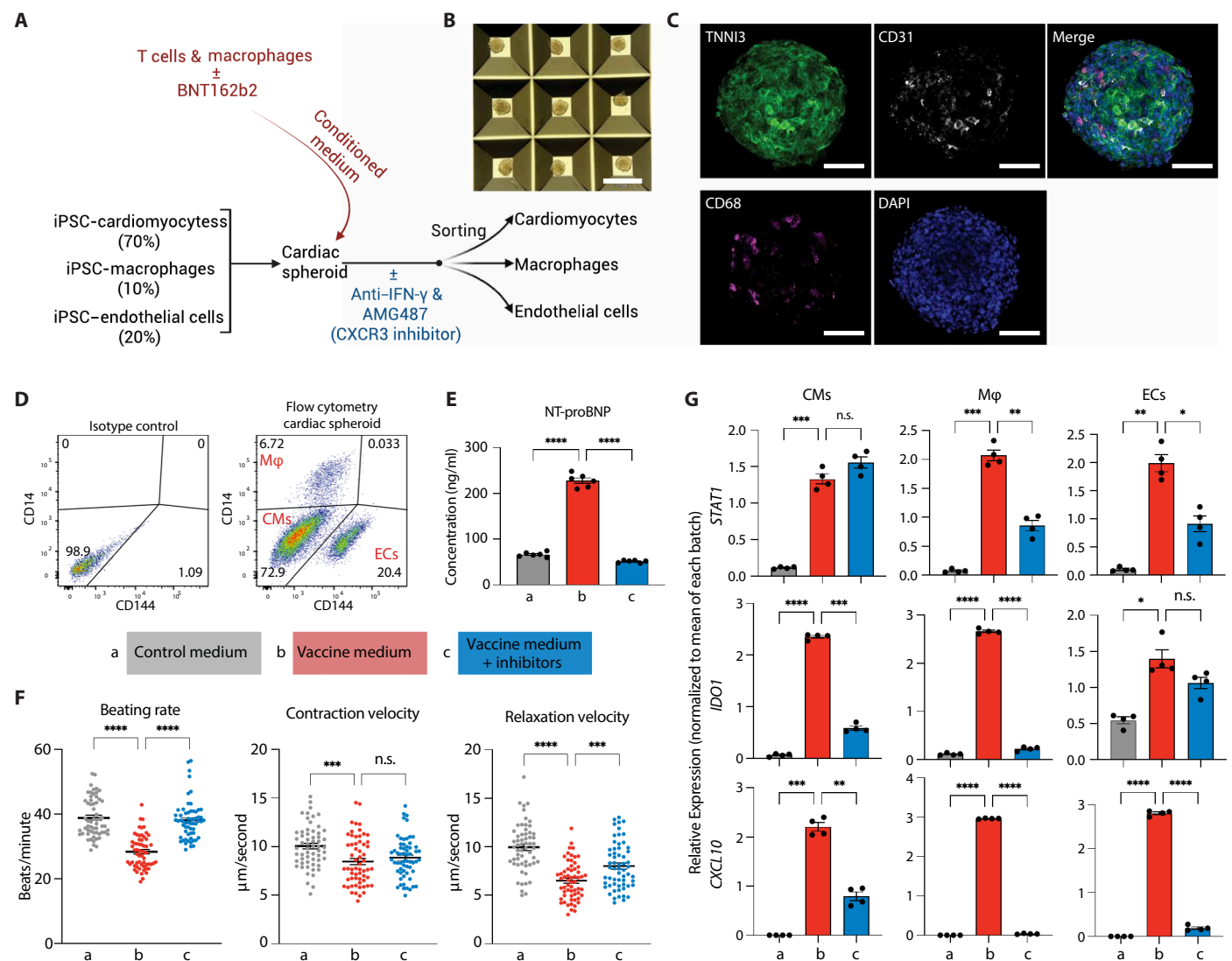


Fig. 3. Targeting CXCL10 and IFN- γ attenuates vaccine-mimicked cardiac injury in spheroids. (A) Schematic of the experimental design. Conditioned medium of immune cell was collected 24 hours after BNT162b2 treatment. iPSC-derived cardiac spheroids were treated with the conditioned medium at a concentration of 30% (v/v) for 24 hours with or without cytokine inhibitors [IFN- γ neutralizing antibody (1 μ g/ml) and CXCL10/CXCR3 inhibitor AMG487 (1 μ M)]. (B) Representative bright-field image of untreated spheroids. Scale bar, 520 μ m. (C) Representative immunofluorescence images of control cardiac spheroids stained for TNNI3 (green, cardiomyocyte marker), CD31 (white, endothelial cell marker), CD68 (magenta, macrophage marker), and 4',6-diamidino-2-phenylindole (DAPI; blue, nuclei). Scale bars, 100 μ m. (D) Flow cytometry of cardiac spheroids stained with isotype controls (left) or with antibodies for macrophage marker CD14 and endothelial marker CD144 (right). M ϕ , macrophages. (E) NT-proBNP secretion from cardiac spheroids treated with conditioned medium from vaccine-stimulated immune cells, with or without cytokine inhibitors (anti-IFN- γ antibody and AMG487) for 24 hours ($n = 6$ independent measurements). (F) Functional analysis of spheroids after 24 hours of treatment, including spontaneous beating rate, contraction velocity, and relaxation velocity ($n = 60$ independent measurements). (G) Expression of IFN- γ -responsive genes in sorted cell populations: cardiomyocytes (CD144⁻/CD14⁻), macrophages (CD144⁻/CD14⁺), and endothelial cells (CD144⁺/CD14⁻). Data were collected from four independent experiments and normalized to the mean of each experiment. For multiple comparisons in (E) to (G), statistical significance was determined using ANOVA with Holm-Sidak's post hoc test. Error bars represent $\pm$ SEM. Statistical significance is defined as not significant (n.s.), * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

before cytokine exposure (Fig. 5A). Quantitative polymerase chain reaction (qPCR) analysis showed that genistein effectively blocked the cytokine-induced up-regulation of *STAT1*, *IDO1*, and *NPPB* (Fig. 5, B to D), and E2, at a physiological concentration, partially mitigated the up-regulation of *STAT1* and *NPPB*. Genistein pretreatment also decreased the release of NT-proBNP (Fig. 5E). These findings highlighted the potential efficacy of genistein in mitigating cytokine-induced damage in cardiomyocytes.

Proteomic analysis at 48 hours post cytokine treatment revealed a decrease in cardiac structural proteins, such as myosin light-chain 12A/B (MYL12A/B), tubulin alpha-1b (TUBA1B), and myosin light-chain kinase 3 (MYLK3), and an increase in inflammatory markers, such as *STAT1* and interferon-induced protein with tetratricopeptide repeats 1 (IFIT1), implying enhanced degradation of structural proteins in the hyperinflammatory conditions (Fig. 5F). Treatment with genistein effectively restored the altered amounts of

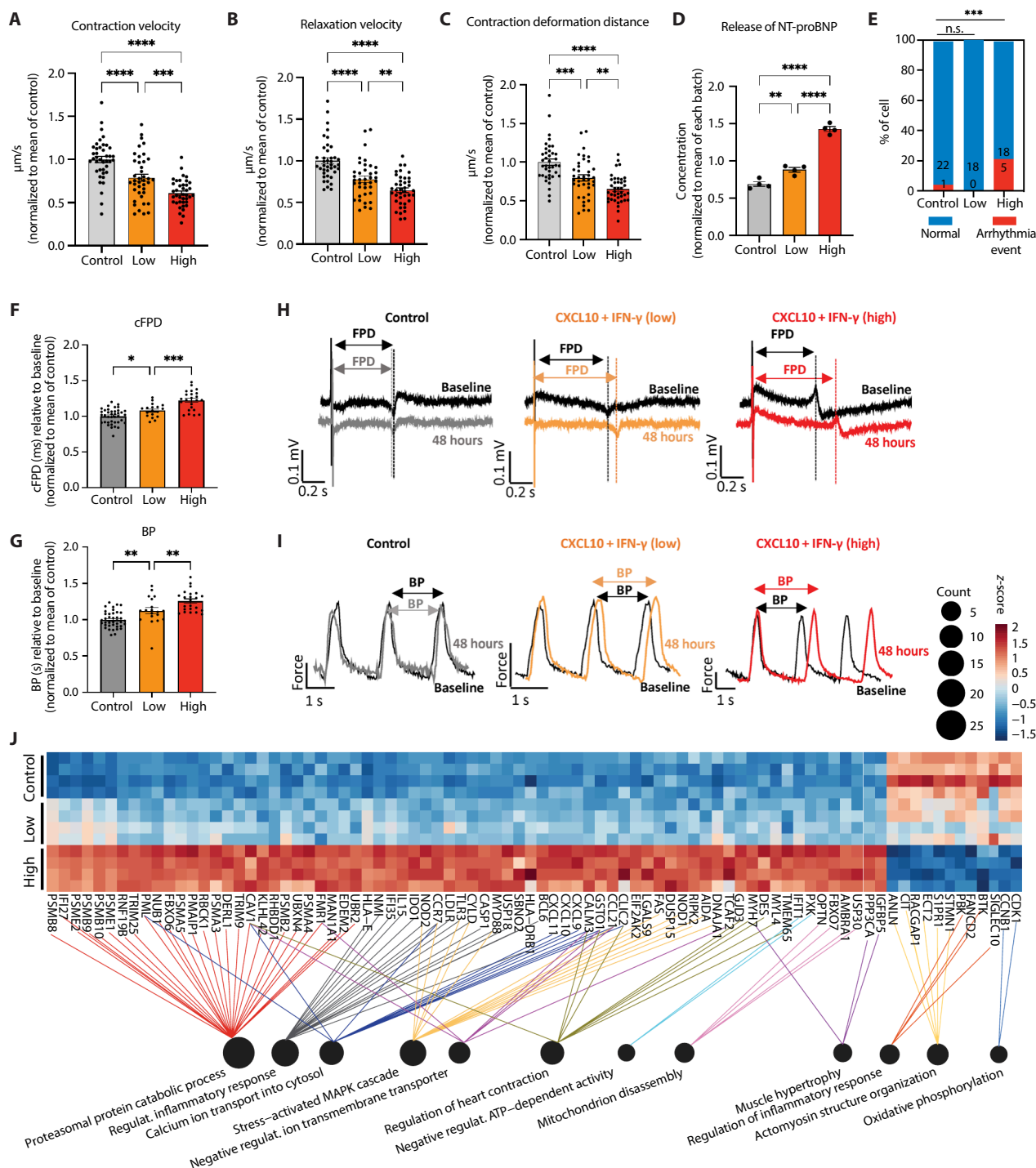


Fig. 4. CXCL10 and IFN- γ directly impair cardiomyocyte function in vitro. (A to C) iPSC-CMs were exposed to the cocktail at low [CXCL10 (1.0 ng/ml) and IFN- γ (0.5 ng/ml)] or high [CXCL10 (10 ng/ml) and IFN- γ (5 ng/ml)] concentrations. Contractility metrics (contraction velocity, relaxation velocity, and deformation distance) measured at 48 hours after treatment (4 iPSC lines $\times$ 10 independent measurements, normalized to the mean of the control within each line). (D) NT-proBNP release in response to cytokine exposure for 48 hours ($n = 4$ iPSC lines). (E) Stacked-column chart showing the percentage of arrhythmia-like events in iPSC-CMs after 48 hours of treatment. Fisher's exact test was used for statistical significance. (F and G) Electrophysiological changes measured by a MEA after 48 hours of treatment: Fridericia's rate-cFPD and BP. A minimum of 16 independent measurements per group were analyzed, and data were normalized to mean of control. (H and I) Representative MEA waveforms of field potential and impedance-based contractility at baseline and after 48 hours of treatment with vehicle (gray), low (orange), and high (red) cytokine concentrations. (J) RNA-seq analysis of cytokine-treated iPSC-CMs from four different iPSC lines. Heatmap illustrating expression patterns of DEGs between high and control groups in representative GOs. $n = 4$ iPSC lines. Color intensity represents the z-score across samples. Complete DEG and GO lists are provided in data files S3 and S4. For multiple comparisons in (A) to (G), statistical significance was determined using ANOVA with Holm-Šidák's post hoc test. Error bars represent $\pm$ SEM. Statistical significance is defined as not significant (n.s.), * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$. ATP, adenosine 5'-triphosphate.

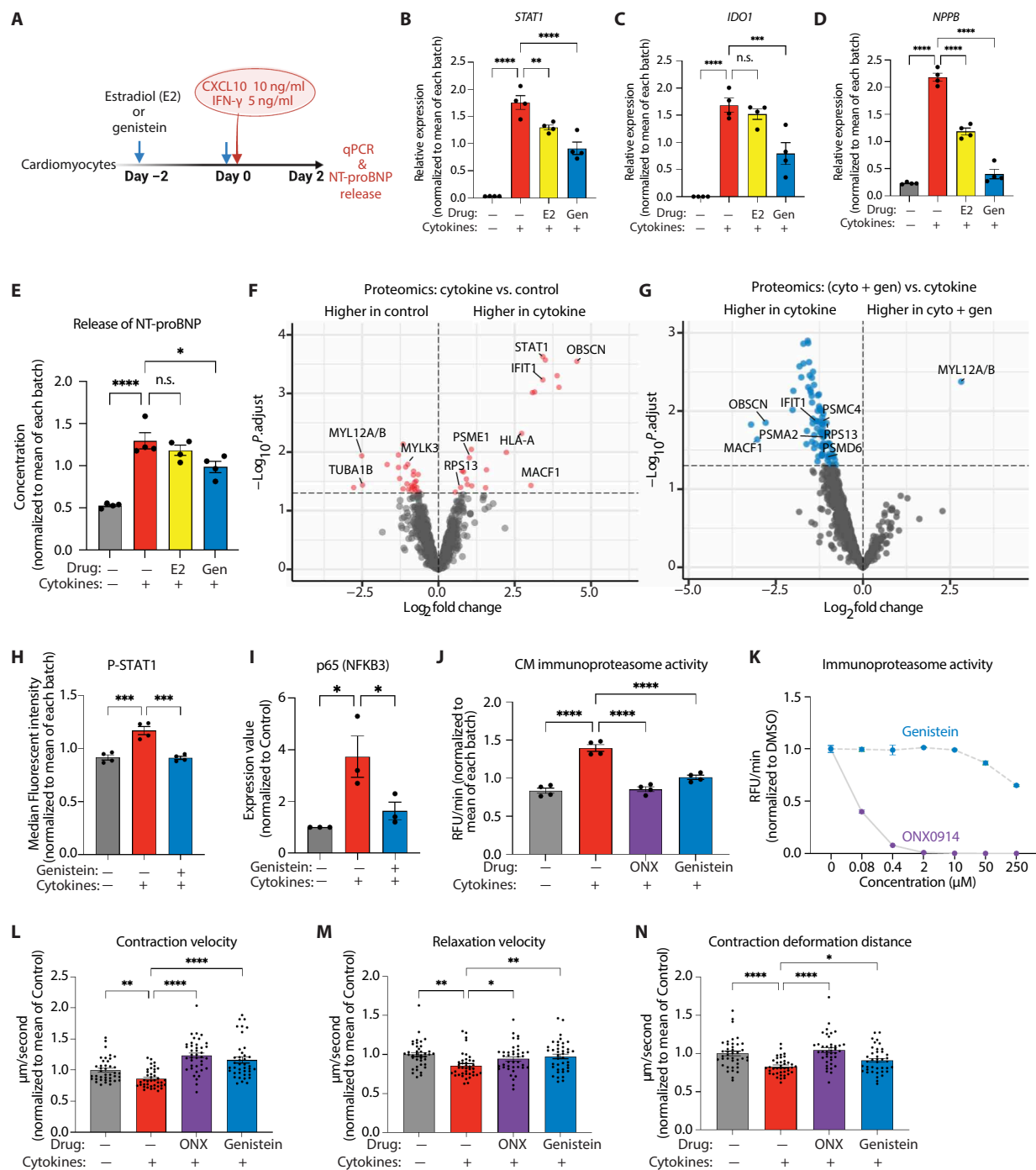


Fig. 5. Genistein protects iPSC-CMs from cytokine-induced injury. (A) Experimental design: iPSC-CMs were pretreated with estradiol (E2, 2 nM) or genistein (10 μ M) for 48 hours before cytokine exposure [CXCL10 (10 ng/ml) and IFN- γ (5 ng/ml)]. (B to D) qPCR quantification of *STAT1*, *IDO1*, and *NPPB* in iPSC-CMs after 48 hours of treatment, normalized to the mean of each batch ($n = 4$ iPSC lines). (E) NT-proBNP release from iPSC-CMs after 48 hours of treatment, normalized to the mean of each batch ($n = 4$ iPSC lines). (F and G) Liquid chromatography–tandem mass spectrometry proteomic analysis of iPSC-CMs showing proteins with significant changes in abundance: control versus cytokine groups (F) and cytokine versus cytokine and genistein (cyto + gen) groups (G) ($n = 3$ iPSC lines). Significant proteins are listed in data file S5 ($P_{\text{adjust}} < 0.05$). (H) Quantification of phosphorylated STAT1 by flow cytometry in iPSC-CMs after 10 min of cytokine treatment ($n = 4$ iPSC lines). (I) NF- κ B P65 (NFKB3) protein quantification by Western blot in iPSC-CMs from three independent experiments, normalized to control ($n = 4$ independent experiments). (J) Immunoproteasome activity in iPSC-CM lysates after 48 hours cytokine treatment with genistein (10 μ M). ONX-0914 (70 nM) was used as a positive control, normalized to the mean of each iPSC line ($n = 4$ iPSC lines). (K) Activity of purified cell-free immunoproteasome in the presence of different concentrations of ONX-0914 or genistein, normalized to DMSO. RFU/min, relative fluorescence units. (L to N) Contractile analysis of iPSC-CMs after 48 hours cytokine treatment, normalized to the mean of control (4 iPSC lines $\times$ 10 independent measurements). For multiple comparisons in (B) to (E), (H) to (J), and (L) to (N), statistical significance was determined using ANOVA with Holm–Šidák’s post hoc test. Error bars represent $\pm$ SEM. Statistical significance is defined as not significant (n.s.), $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$.

structural and inflammatory proteins such as MYL12A/B, obscurin (OBSCN), microtubule actin cross-linking factor 1 (MACF1), and IFIT1 to amounts comparable to controls (Fig. 5G and fig. S11B). Additionally, phosphorylation of STAT1 (a driver of IFN- γ -responsive genes) and up-regulation of nuclear factor κ B p65 subunit (NF- κ B P65 or NFKB3) were mitigated by genistein treatment, confirming the anti-inflammatory effects of genistein in cytokine-treated iPSC-CMs (Fig. 5, H and I; and fig. S11, C and D).

Given that both RNA-seq and proteomics data indicated an activation of the immunoproteasome in cytokine-treated iPSC-CMs, potentially leading to the degradation of cardiac structural proteins (Figs. 4J and 5F and figs. S10I and fig. S11A), we further investigated whether genistein exerted its cardiac protective effects through the immunoproteasome pathway. Cytokine treatment increased immunoproteasome activity in iPSC-CMs, which was fully blocked by pretreatment with the immunoproteasome-specific inhibitor ONX-0914, whereas the constitutive proteasome inhibitor MLN2238 and nonspecific proteasome inhibitor PS341 only partially inhibited the activity (fig. S11E). Genistein blocked immunoproteasome activity in cytokine-treated iPSC-CMs, similar to ONX-0914 (Fig. 5J). Genistein showed minimal inhibitory effect on purified cell-free immunoproteasome, suggesting that its inhibitory effects might be mediated through upstream pathways rather than direct binding (Fig. 5K). Both ONX-0914 and genistein effectively mitigated cytokine-induced contractile defects (Fig. 5, L to N), underscoring the role of the immunoproteasome in cytokine-induced cardiac injury.

In addition, we investigated the effects of CXCL10 and IFN- γ , as well as the protective role of genistein, in other cell types. In iPSC-ECs, pretreatment with genistein reduced the cytokine-induced, dose-dependent up-regulation of inflammatory genes, such as *IDO1*, *TNF*, and *IL1B* (fig. S12, A and B), and diminished the increase in cell-surface ICAM-1, which mediates adhesion and recruitment of leukocytes to inflammatory sites (fig. S12C). Consistently, cell adhesion assays showed that cytokine-induced THP-1 adhesion to endothelial cells was mitigated by genistein (fig. S12, D and E). Similarly, in iPSC-MACs, genistein effectively reduced the release of CXCL10 and IFN- γ (fig. S13, A and B) and the expression of IFN- γ -responsive and inflammatory genes induced by vaccine treatment (fig. S13C). Genistein also mitigated the release of the two cytokines from human PBMCs (fig. S13, D and E). These results suggest that genistein has a protective effect not only in reducing direct cardiomyocyte injury induced by cytokines but also in mitigating inflammation in endothelial and immune cells.

Genistein mitigates cytokine-induced myocardial injury in vivo

To investigate whether a transient surge of the two cytokines, CXCL10 and IFN- γ , is sufficient to trigger a myocardial injury in vivo, we first conducted a pilot experiment in wild-type BALB/c mice. Mice received a one-time intravenous injection of recombinant mouse CXCL10 (0.1 mg/kg) and IFN- γ (0.1 mg/kg). Because the onset of vaccine-associated myocarditis in patients (1 to 3 days postvaccination) coincides with the surge in IFN- γ (1 day postvaccination) (22), we monitored time-course changes of cardiac injury marker cTnI on the day of cytokine administration. A single injection of the cytokine cocktail led to an increase in serum cTnI with the peak occurring at 4 hours postinjection (fig. S14, A to C). Accordingly, blood and tissue samples were collected 4 hours postinjection in the subsequent experiments (Fig. 6A).

Next, mice were assigned to three groups: control, cytokine, and cytokine plus genistein, where mice received genistein (50 mg/kg) pretreatment or vehicle (corn oil) by oral gavage for 7 days before the intravenous administration of the cytokine cocktail or vehicle (PBS). The regimen of genistein treatment was determined on the basis of our previous study (17). The serum cTnI amount elevated by cytokine injection was reduced by genistein treatment, indicating the mitigative effect of genistein (Fig. 6B). Immunofluorescence staining revealed that the increased infiltration of LY6G⁺ neutrophils and CD68⁺ macrophages in the hearts of the cytokine-treated mice was effectively reduced by genistein (Fig. 6, C to F). Furthermore, genistein reduced the cytokine-induced immunoproteasome activity in the heart, confirming its mitigative effect on immunoproteasome activity in vivo (Fig. 6G). Transcriptomic analysis identified 152 Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways enriched in cytokine-treated mice compared with controls, predominantly related to inflammatory responses (Fig. 6H and fig. S14, D and E), and genistein pretreatment down-regulated 30 of these pathways, including pathways related to glycolysis, reactive oxygen species responses, oxidative phosphorylation, and diabetic cardiomyopathy (Fig. 6I and fig. S14F).

Genistein mitigates vaccine-induced myocardial injury in vivo

Last, we examined whether genistein could prevent mRNA vaccine-induced cardiac injury in vivo. Mice were divided into three groups: control, vaccine, and vaccine plus genistein. Mice in the vaccine groups received BNT162b2 on days 0 and 21 [2 μ g intramuscular (im), each], whereas control mice received PBS (im). Genistein (50 mg/kg) or vehicle (corn oil) was administered by oral gavage for 7 days before the second dose (Fig. 7A).

Vaccination markedly increased the circulating cytokines CXCL10 and IFN- γ , and this rise was blunted by genistein (Fig. 7B). Genistein did not diminish vaccine immunogenicity. Anti-SARS-CoV-2 Spike IgG titers in genistein-treated mice were comparable to those in vaccine-only animals (Fig. 7C). By contrast, genistein significantly reduced the vaccine-induced elevation of serum cTnI (Fig. 7D). Correspondingly, up-regulated expression of IFN- γ -responsive genes (*Stat1* and *Ido1*) and leukocyte markers (*Ptprc* and *Ly6g*) in the heart was ameliorated in the genistein group (Fig. 7E). Similar up-regulation of inflammatory genes by vaccination was observed in other organs, including the lung, liver, and kidney, which was rescued by either CXCL10/IFN- γ blockade or genistein treatment (fig. S15, A and B). Immunohistochemistry further confirmed that vaccination-induced neutrophil and macrophage infiltration was reduced by genistein treatment (Fig. 7, F to I). These results indicate that oral administration of genistein could mitigate the CXCL10/IFN- γ surge and subsequent myocardial injury induced by mRNA vaccination, with preserved humoral immunity in vivo.

DISCUSSION

Although vaccine-associated myocarditis is rare, it has emerged as a notable safety concern, given that over 5 billion people have been vaccinated globally. The infrequent nature of this adverse effect complicates the accessibility of patient samples and the development of suitable animal models, thereby challenging our understanding of its underlying molecular mechanisms (19). To overcome these challenges, we integrated published proteomic data from

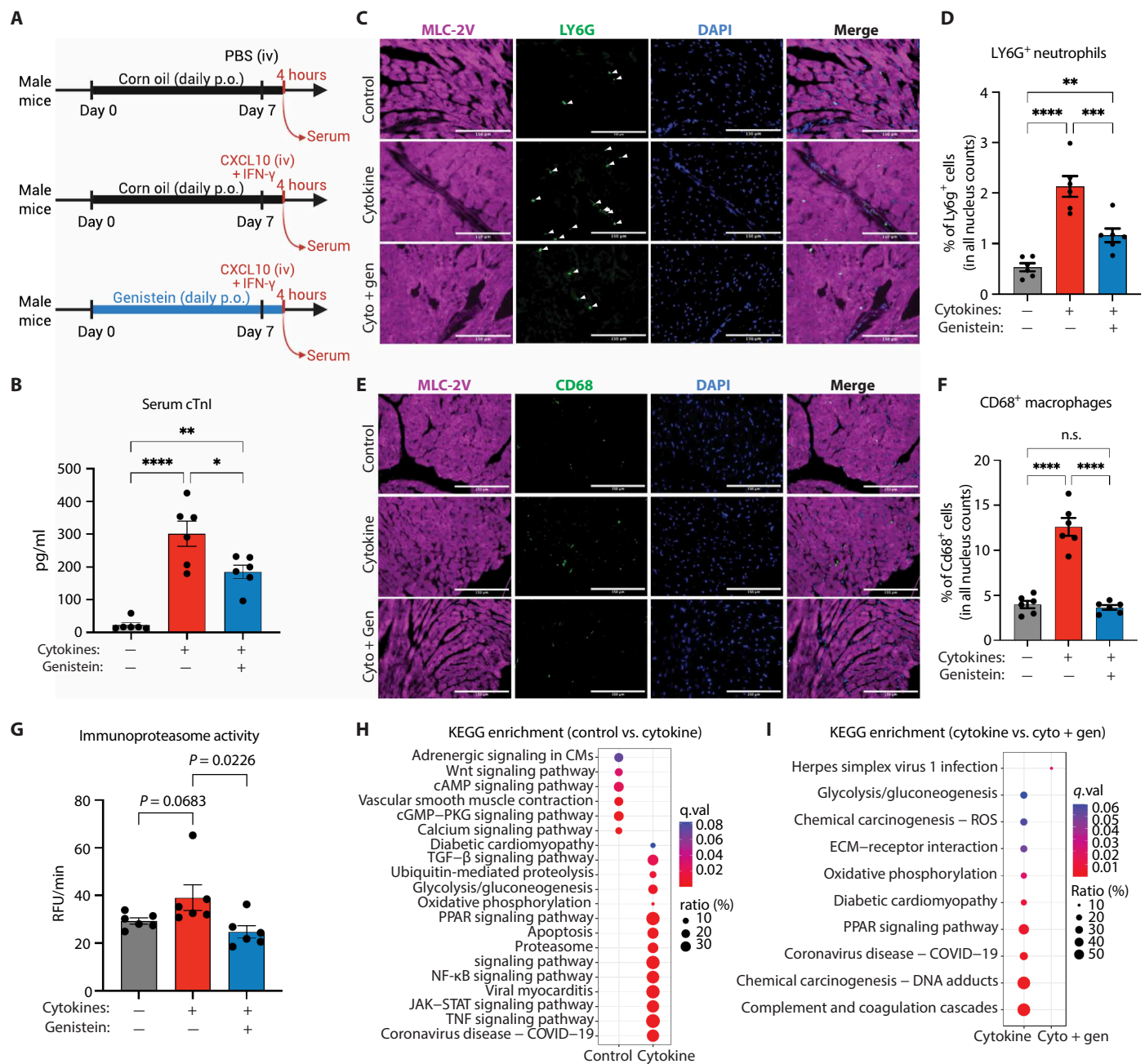


Fig. 6. Genistein mitigates cytokine-induced myocardial injury in mice. (A) Experimental design: wild-type BALB/c mice pretreated with genistein [50 mg/kg, oral gavage (p.o.), 7 days] before intravenous (iv) injection of CXCL10 and IFN- γ (0.1 mg/kg each). (B) Serum cTnI amounts 4 hours after cytokine injection ($n = 6$ mice per group). (C) Immunofluorescence staining showing neutrophil infiltration in hearts. Heart tissue sections were stained for MLC-2V (magenta, cardiomyocyte marker), LY6G (green, neutrophil marker), and DAPI (blue, nuclei). Scale bars, 150 μ m. (D) Quantification of LY6G $^{+}$ cells in mouse heart sections. $n = 6$ mice per group. (E) Immunofluorescence staining showing macrophage infiltration in hearts. Heart tissue sections were stained for MLC-2V (magenta, cardiomyocyte marker), CD68 (green, macrophage marker), and DAPI (blue, nuclei). Scale bars, 150 μ m. (F) Quantification of CD68 $^{+}$ cells in mouse heart sections. $n = 6$ mice per group. (G) Immunoproteasome activity in heart tissue lysates ($n = 6$ mice per group). (H and I) KEGG pathway analysis from bulk RNA-seq of heart tissues ($n = 6$ mice per group): control versus cytokine (H) and cytokine versus cytokine and genistein (I). A complete list of affected pathways is provided in data file S6. For multiple comparisons in (B), (D), (F), and (G), statistical significance was determined using ANOVA with Holm-Šidák's post hoc test. Error bars represent $\pm$ SEM. Statistical significance is defined as not significant (n.s.), $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$. cAMP, cyclic adenosine 3',5'-monophosphate; PKG, cGMP (cyclic guanosine 3',5'-monophosphate)-dependent protein kinase; TGF- β , transforming growth factor- β ; PPAR, peroxisome proliferator-activated receptor; ECM, extracellular matrix; ROS, reactive oxygen species.

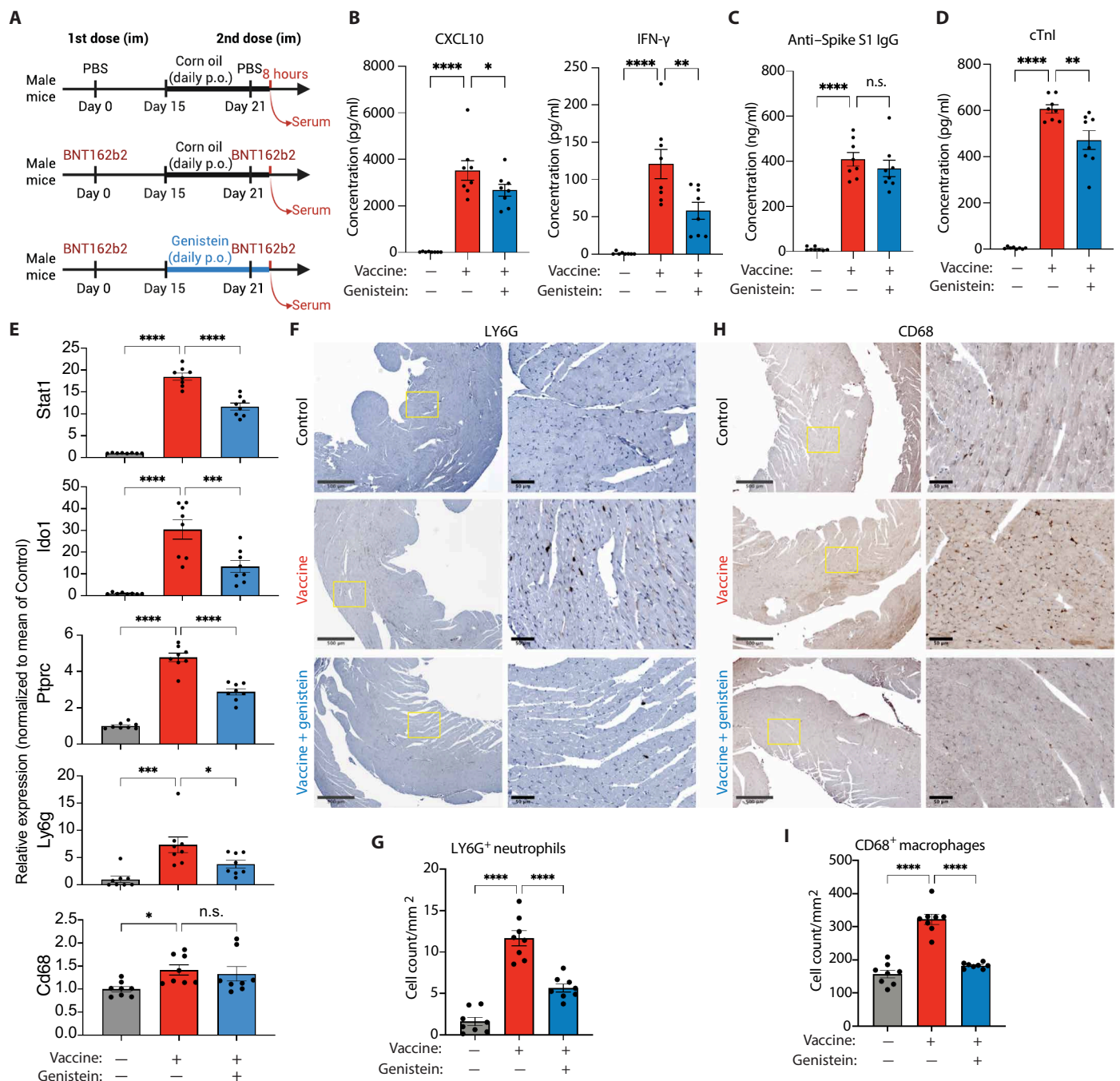


Fig. 7. Genistein mitigates mRNA vaccine-induced myocardial injury in mice. (A) Experimental timeline: BALB/c mice received PBS or BNT162b2 (2 µg, im, days 0 and 21) with or without genistein pretreatment (50 mg/kg, oral gavage, 7 days before second dose). (B) Serum CXCL10 and IFN-γ concentrations after second vaccination. *n* = 8 mice per group. (C) Serum anti-COVID-19 Spike S1 IgG concentration by enzyme-linked immunosorbent assay. *n* = 8 mice per group. (D) Serum cTnI concentrations after second vaccination. *n* = 8 mice per group. (E) qPCR analysis of IFN-γ-responsive genes (*Stat1* and *Ido1*) and leukocyte markers (*Ptprc*, *Ly6g*, and *Cd68*) in heart tissues after second vaccination (normalized to control). *n* = 8 mice per group. (F) Immunohistochemistry showing LY6G⁺ neutrophil infiltration in vaccinated hearts. Enlarged insets from representative areas (yellow boxes) are shown at right. Scale bars, 500 µm (left) and 50 µm (right). (G) Quantification of LY6G⁺ cells. Quantification is based on 10 field of views for each section per mouse (*n* = 8 mice per group). (H) Immunohistochemistry showing CD68⁺ macrophage infiltration in vaccinated hearts. Enlarged insets from representative areas (yellow boxes) are shown at right. Scale bars, 500 µm (left) and 50 µm (right). (I) Quantification of CD68⁺ cells. Quantification is based on 10 field of views for each section per mouse (*n* = 8 mice per group). For multiple comparisons in (B) to (E), (G), and (I), statistical significance was determined using ANOVA with Holm-Šidák's post hoc test. Error bars represent ± SEM. Statistical significance is defined as not significant (n.s.), **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001.

vaccinated individuals with in vitro human iPSC models and in vivo mouse models. Both human proteomic and iPSC-MAC data showed up-regulation of CXCL10 and IFN- γ after vaccination, consistent with findings from an independent cohort of patients with myocarditis (20). Using human iPSC-CM, cardiac spheroid, and mouse models, we then confirmed that these two cytokines are key drivers of myocardial injury. Our findings across these models support a two-phase framework. In the first initiation phase, innate sensing of vaccine mRNA activates tissue macrophages, which secrete CXCL10 while producing minimal IFN- γ . In the second amplification phase, CXCL10 recruits and primes T cells, which then release IFN- γ . The resulting rise in circulating CXCL10 and IFN- γ , as demonstrated in our models, directly injures cardiomyocytes and promotes immune-cell infiltration (fig. S16). This framework is supported by a recent scRNA-seq study of myocarditis biopsies, which identified *IL18* (a macrophage-derived IFN- γ inducer) as a hallmark gene of postvaccination myocarditis (32), consistent with the macrophage-T cell interactions observed in our models.

Although IFN- γ plays a pivotal role in the body's defense against viral infections, including its protective effects in typical viral myocarditis, our findings indicate that this signaling has a context-dependent impact. In the setting of mRNA COVID-19 vaccination, up-regulated IFN- γ signaling appears to contribute to myocardial injury, distinguishing vaccine-associated myocardial injury from typical viral myocarditis. This observation is consistent with prior studies showing the detrimental impact of elevated IFN- γ signaling in cardiac disease models (33–37). For example, in a study of transgenic mice with constitutive overexpression of IFN- γ in the livers (35), elevated serum amounts of IFN- γ led to myocarditis-like phenotypes in the heart, such as myofiber atrophy and dilated cardiomyopathy. In another mechanistic study, primary rat cardiomyocytes exposed to IFN- γ led to the degradation of myosin heavy-chain proteins through the ubiquitin-proteasome system (37). Moreover, IFN- γ and its downstream Janus kinase (JAK)/STAT pathways have been implicated in cardiac injury (38–40) and proatherogenic inflammatory responses (41) in COVID-19. For instance, in human cardiac organoids (39), a cocktail of IFN- γ , IL-1 β , and poly(I:C) recapitulated cytokine storm-induced cardiac injury observed in severe COVID-19 cases. Our results further confirm the cardiotoxic properties of the IFN- γ signaling pathway, adding the new insight that the CXCL10-IFN- γ axis is involved in the development of myocardial injury post mRNA vaccination. Of note, TNF- α , which is one of the most well-studied proinflammatory cytokines and implicated in viral myocarditis and chronic heart failure (42), was not up-regulated in postvaccination human plasma or in the supernatant of iPSC-MACs in vitro, suggesting that myocardial injury after mRNA vaccination is mediated by selective up-regulation of IFN- γ signaling rather than by generalized systemic inflammation.

Given that our proposed mechanism involves a surge of circulating cytokines, it is reasonable to believe that the inflammatory response may extend to other organs. Transcript profiling showed that mRNA vaccination up-regulates IFN- γ -responsive genes and immune-cell markers in the lung, liver, and kidney. Neutralization of CXCL10/IFN- γ , as well as genistein treatment, reversed these changes, indicating that CXCL10-IFN- γ signaling is a key driver of a systemic inflammatory response after mRNA vaccination. These results align with clinical reports showing postvaccination inflammation in non-cardiac organs (43–45), although myocarditis remains the most frequently recognized manifestation. Two factors may account for this

observation. First, cardiac injury typically presents with acute chest pain and can be confirmed rapidly with widely available tests such as cTnI, creatine kinase-MB (CK-MB), and EKG. These features, especially when observed in adolescents, prompt immediate evaluation and facilitate diagnosis. In contrast, inflammation of other organs often produces nonspecific symptoms and may require advanced imaging modalities such as contrast computed tomography (CT) and MRI for detection, thereby raising the diagnostic threshold. Second, cardiomyocytes might be intrinsically more sensitive to CXCL10/IFN- γ signaling than other cell types, although this possibility remains to be investigated. Determining whether detection bias, tissue-specific susceptibility, or a combination of these and other factors account for the clinical predominance of cardiac injury is an important objective for future studies.

Our results indicate that the activation of the immunoproteasome plays a role in vaccine-induced myocardial injury by degrading cardiac sarcomere proteins. Specifically, cytokine-treated iPSC-CMs exhibited altered gene-expression patterns akin to myocarditis, with up-regulation of the proteasome pathway. This aberrant activation was effectively counteracted by both ONX-0914 (an immunoproteasome-specific inhibitor) and genistein, which restored contractile function and normalized immunoproteasome activity. These findings are consistent with previous reports demonstrating that ONX-0914 protects against acute viral myocarditis in mice (46), reinforcing the therapeutic potential of targeting activated proteasomes. Our data suggest that genistein's mitigative effect on immunoproteasome activity is indirect, likely mediated through upstream targets or its pleiotropic effects, distinguishing it from the direct action of ONX-0914.

To explore potential solutions, we used the clinical observation of male predominance in vaccine-associated myocarditis as a conceptual clue to test an estrogen-mimetic strategy, based on the possibility that estrogen signaling might confer protection. We first tested estradiol (E2) and the phytoestrogen genistein (47, 48) in vitro, and both showed protective effects against cytokine-induced cardiomyocyte injury at the level of gene-expression changes. For subsequent in vivo studies, we selected genistein as a more translationally feasible approach, given its oral bioavailability and favorable safety record as a soy-derived dietary supplement with anti-inflammatory properties (17), whereas direct estrogen therapy would not be practical as a preventive intervention. Our experiments with genistein were not designed to explain the clinical male predominance of vaccine-associated myocarditis but rather to evaluate a pharmacologic intervention in preclinical models. Thus, our findings provide proof of concept that genistein can attenuate cytokine-driven myocardial injury, whereas the biological basis of sex bias remains an important area for future investigation.

One important aspect of this study is that myocardial injury by elevated IFN- γ signaling can be a class effect of mRNA vaccines, given that IFN- γ signaling is a fundamental defense mechanism against exogenous DNA and RNA molecules, including viral nucleic acids. Excess activation of IFN- γ signaling could lead to adverse outcomes when new mRNA vaccines are developed for infectious diseases and malignancies in the future. Therefore, the potential adverse events driven by excess circulating cytokines should be carefully considered, and pretreatment with anti-inflammatory agents such as genistein may offer a preventive strategy for those at higher risk.

Several limitations of our study should be acknowledged. First, because the incidence of vaccine-associated myocarditis is very low,

creating an animal model that fully reproduces the clinical presentation is challenging. To unmask the phenotype, we used a supraphysiological dose of BNT162b2 (2 μ g per injection) rather than the body-surface-area-adjusted human-equivalent dose (0.2 μ g). This 2- μ g regimen reliably produced cardiac injury and preserved the second-dose predominance seen in patients, whereas the 2- μ g dose remained lower than the 5- μ g dose used in a previous study (21). Our optimized protocol, which used young male mice, intramuscular administration (the clinical route), and sampling at the peak-injury time point, enabled us to detect cardiac injury that earlier mouse studies observed only minimally or under additional stresses (49–51). Our model also recapitulated immune cell infiltration, consistent with recent scRNA-seq data from biopsy samples of postvaccination myocarditis (32). However, the contrast between first- and second-dose injury in mice was less pronounced than in humans, and the 2- μ g model did not replicate the clinical sex bias. These limitations likely reflect the high dose that we used and the lack of genetic diversity in laboratory mice, underscoring the need for more refined models. Future work could combine human-equivalent dosing with genetic sensitization (for example, by introducing human susceptibility alleles) to better replicate clearer second-dose predominance and sex-specific risk. Second, the impact of CXCL10/IFN- γ and their inhibition on the immune system and noncardiac tissues need further investigation. Although CXCL10/IFN- γ neutralization largely preserved systemic cytokine responses, we observed a partial reduction in CCL2 and anti-Spike IgG titers. We cannot exclude the possibility that broader or longer-term aspects of vaccine immunity might be blunted by cytokine blockade. Besides, follow-up studies that assess the direct impact of CXCL10/IFN- γ on the heart (for example, by using cardiac-specific receptor knockout) will also be needed. Third, our targeted cytokine arrays might have missed additional mediators. Last, our study did not encompass other potential mechanisms such as autoantibodies against cardiac proteins (52) and IL-1 receptor antagonist (53).

In conclusion, our study implicates the CXCL10–IFN- γ axis as a key mediator of myocardial injury in multiple preclinical models of mRNA vaccination and proposes a potential strategy to mitigate this adverse effect. As mRNA technology continues to evolve, mechanistic insights such as those presented here will be crucial in ensuring its safe and effective application across broad therapeutic areas, from pandemic response to cancer treatment and beyond.

MATERIALS AND METHODS

Study design

This study investigated the mechanisms of mRNA vaccine-associated myocarditis and tested whether the anti-inflammatory agent genistein could mitigate this effect. We hypothesized that cytokine surges in the circulation after vaccination injure cardiomyocytes. We combined reanalysis of published human plasma datasets with *in vitro* iPSC models (macrophages, cardiomyocytes, endothelial cells, and three-dimensional cardiac spheroids) and *in vivo* mouse vaccination and cytokine-challenge studies. Primary end points in mice were serum cTnI, cytokine amounts, immune-cell infiltration, and gene expression. *In vitro* end points included cytokine secretion, gene expression, and cardiomyocyte contractility. Sample sizes were guided by preliminary data: at least three independent iPSC lines for cellular assays and eight mice per experimental group, unless otherwise specified. Outliers were excluded only if attributable to

technical failure. Exact sample sizes and statistical tests are detailed in the figure legends and the Supplementary Materials. Mice were randomized to treatment groups, and investigators were blinded for histological and functional analyses. All iPSC lines were obtained from the Stanford Cardiovascular Institute iPSC biobank with informed consent and Institutional Review Board approval, and all animal experiments were performed under Stanford Institutional Animal Care and Use Committee (IACUC)–approved protocols (IACUC-34047).

Human plasma data

Published proteomics datasets from vaccinated individuals (22) and patients with vaccine-associated myocarditis (19) were reanalyzed to identify cytokines elevated after vaccination. Candidate cytokines were prioritized on the basis of temporal induction after vaccination and selective elevation in myocarditis cases.

iPSC-derived models

Human iPSCs from the Stanford Cardiovascular Institute Biobank were differentiated into macrophages (iPSC-MACs), cardiomyocytes (iPSC-CMs), and endothelial cells (iPSC-ECs). iPSC-MACs were treated with BNT162b2 (0.6 to 3 μ g/ml) or mRNA-1273 (2 to 10 μ g/ml) for 24 hours to assess their transcriptional and cytokine responses to mRNA vaccines. iPSC-CMs were exposed to a cytokine cocktail containing CXCL10 and IFN- γ at low (1.0 and 0.5 ng/ml, respectively) or high (10 and 5 ng/ml, respectively) concentrations for 48 hours to evaluate cytokine-induced effects. To assess the effect of genistein *in vitro*, iPSC-MACs, iPSC-CMs, and iPSC-ECs were pretreated with genistein (10 μ M) for 48 hours before cytokine treatment (for iPSC-CMs and iPSC-ECs) or vaccine exposure (iPSC-MACs). Three-dimensional cardiac spheroids comprising iPSC-CMs, iPSC-ECs, and iPSC-MACs mixed at a 7:2:1 ratio were used to evaluate multicellular responses. Further details are provided in the Supplementary Materials.

mRNA vaccine-induced myocardial injury in mice

Male BALB/c or C57BL/6NCr mice (6 to 8 weeks old at purchase, Charles River Laboratories) were used for vaccination experiments. Female mice of the same strains were used only for the sex-bias experiment. BNT162b2 was administered intramuscularly into the left quadriceps under isoflurane anesthesia on day 0 (first) and day 21 (second) at doses of 0.2, 2, or 5 μ g, as specified in Results and the figure legends. On the basis of pilot data identifying 6 to 10 hours postvaccination as the peak injury window (peak cTnI increase), blood and tissue were collected 8 hours after the second dose. In the CXCL10/IFN- γ blocking study, neutralizing antibodies (0.5 mg per mouse, intraperitoneal) were administered 8 hours before the second dose. For the genistein study, mice received genistein (50 mg/kg per day in corn oil, oral gavage) or vehicle (corn oil) for 1 week before the second vaccine dose. Further details are provided in the Supplementary Materials.

Cytokine-induced myocardial injury in mice

Male BALB/c mice received a single tail-vein injection of a cytokine cocktail containing CXCL10 (0.1 mg/kg) and IFN- γ (0.1 mg/kg). Blood and tissue were collected at 4 hours postinjection, corresponding to the peak increase in serum cTnI, unless otherwise indicated. For the genistein study, mice received genistein (50 mg/kg per day in corn oil, oral gavage) or vehicle (corn oil) for 1 week before

cytokine treatment. Further details are provided in the Supplemental Materials.

Statistical analysis

Multigroup comparisons used one-way analysis of variance (ANOVA) with Holm-Šidák's post hoc testing using Prism 10 (GraphPad Software). $P < 0.05$ was considered significant. Biological replicates refer to independent iPSC lines or animals; technical replicates were included as appropriate. Full details of sample size and analyses are in figure legends and Supplementary Materials. Tabulated individual-level data are shown in data file S9.

Supplementary Materials

The PDF file includes:

Materials and Methods

Figs. S1 to S16

Legends for data files S1 to S9

Legend for movie S1

References (54, 55)

Other Supplementary Material for this manuscript includes the following:

Data files S1 to S9

Movie S1

MDAR Reproducibility Checklist

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Acknowledgments: We thank A. Barmada and C. L. Lucas (Yale University) for providing metadata of scRNA-seq of PBMCs from patients with myocarditis. We extend our gratitude to B. Pulendran (Stanford University) for providing invaluable feedback and expert opinions on immunology. We also thank the members of the Wu lab for discussion and feedback, D. Dunham for dedicated effort in managing vaccine samples, and B. Wu for proofreading the manuscript. **Funding:** We are grateful for support by the National Institutes of Health (NIH) R01 HL113006, HL141371, R01 HL141851, R01 HL163680, and R01 HL176822; Gootter-Jensen Foundation (to J.C.W.); Translational Research Institute for Space Health (TRISH) postdoctoral fellowship NNX16A069A; NIH K99/R00 Pathway to Independence Award HL177331 (to X.C.); NIH K99/R00 Pathway to Independence Award HL166773 (to M.N.); NIH K99/R00 Pathway to Independence Award HL179400 (to A.C.); American Heart Association (AHA) Career Development Award 25CDA1450742; AHA postdoctoral fellowship 22POST916641; Tobacco-Related Disease Research Program T32FT4798 (to L.L.); and AHA postdoctoral fellowship 23POST1020812 and grants 25CDA1456151 (to A.M.) and AHA 25POST1372048 (to W.L.); the Henry Gustav Floren Trust; the Stanford Department of Medicine Team Science Program; the Stanford Medicine Office of the Dean; and NIH R01 AI182319. **Author contributions:** X.C. and M.N. conceptualized and designed the project, performed the experiments, analyzed and interpreted the data, and wrote the manuscript. A.M., M.N., and W.L. performed animal experiments. Y.-i.C., X.K., and L.L. performed Western blot and qPCR analysis. A.C. performed proteomics experiments. G.M.-P. performed electrophysiological assays. X.C. and Wenjuan Zhu performed transcriptomic analysis. Wenshu Zeng performed immunofluorescence staining. S.R.Z. and J.W.S.J. performed cardiomyocyte differentiation. P.J.U. and K.C.N. conceptualized the project and provided expert opinions on immunology. J.C.W. conceptualized and designed the project, analyzed and interpreted the data, provided financial support, and approved the final manuscript. **Competing interests:** J.C.W. is a cofounder and scientific advisory board member of Greenstone Biosciences. All other authors declare that they have no competing interests. **Data and materials availability:** All data associated with this study are present in the paper or the Supplementary Materials. The RNA-seq data and proteomics data are available at Gene Expression Omnibus (GEO) (accession GSE254539) and ProteomeXchange (accession PXD049000), respectively. All human iPSC lines used in this study will be available upon reasonable request with a material transfer agreement (MTA).

Submitted 24 April 2024

Resubmitted 22 July 2025

Accepted 18 November 2025

Published 10 December 2025

10.1126/scitranslmed.adq0143